

A zero-level boundary current from holomorphic BF theory

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Holomorphic BF theory and a level-zero current

A current algebra can lose its central term while its bulk field theory retains a nondegenerate Batalin–Vilkovisky pairing. The mechanism is a change in the differential, with two field components kept separate. On a complex curve times a real half-line, the quadratic action density is

$$\mathcal{L}_\kappa(a, b) = b \wedge d' a + \frac{\kappa}{2} a \wedge \partial a, \quad d' = \bar{\partial} + d_t.$$

It is defined for smooth fields. On compactly supported fields the action is $S_\kappa(a, b) = \int_M^{\text{res}} \mathcal{L}_\kappa(a, b)$. Here a is a partial connection in the antiholomorphic and real directions, and b is a holomorphic one-form. The pairing between these fields is independent of κ . At $\kappa = 0$ the action is holomorphic BF theory: the field b multiplies the partial curvature $d' a$. Its chiral boundary current has zero singular operator product, but its polynomial fields and all holomorphic jets remain. A compactly supported collar map identifies the quantum boundary observables with the boundary value of the bulk theory.

1.1 THE FIELDS, INCLUDING THEIR ODD COMPONENTS

Let $\Sigma = \mathbb{C}$ with coordinate $z = x + iy$, and let $M = \Sigma \times \mathbb{R}_{\geq 0}$ with coordinate t on the second factor. Choose the orientation whose induced boundary orientation on $\Sigma \times \{0\}$ is the complex orientation. Write $\mathcal{A}$ for smooth forms generated by $d\bar{z}, dt$. All coefficients are complex-valued, and

$$d' = d\bar{z} \wedge \partial_{\bar{z}} + dt \wedge \partial_t, \quad \partial = dz \wedge \partial_z.$$

These operators satisfy $(d')^2 = \partial^2 = d' \partial + \partial d' = 0$. The parameter $\kappa \in \mathbb{C}$ is a scalar of cohomological degree zero. Statements polynomial in κ also hold over $\mathbb{C}[\kappa]$.

The coefficient space of α is $\mathbb{C}e$, and that of β is its dual $\mathbb{C}e^\vee$. These chosen bases suppress the evaluation pairing in the scalar formulas. The fields have two summands,

$$E_\kappa = \mathcal{A}[1] \oplus (\mathcal{A} \wedge dz)[1].$$

This is the unrestricted smooth field complex. Write $E_{\kappa,c}$ for its subcomplex of compactly supported fields; compact support is required in both the curve and the real direction and may meet the boundary of M . The notation records the following precise degrees: an ordinary p -form in either summand has cohomological degree $p - 1$. The differential is the actual exterior operator

$$Q_\kappa(\alpha, \beta) = (d' \alpha, d' \beta + \kappa \partial \alpha).$$

Thus the holomorphic one-form β has degree zero before adding any $d\bar{z}$ or dt . All tensor substitutions use the Koszul sign of the displayed cohomological degrees.

For a component of the field complex, the corresponding coordinate has the opposite degree. This coordinate degree is the ghost number. The complete list is

component	form degree	ghost number	role
$\alpha^{(0)}$	0	1	gauge ghost c
$\alpha^{(1)} = a$	1	0	partial connection
$\beta^{(1)} = b$	1	0	holomorphic one-form
$\alpha^{(2)}$	2	-1	antifield of b
$\beta^{(2)}$	2	-1	antifields of a
$\beta^{(3)}$	3	-2	antifield of c .

There is no $\alpha^{(3)}$, because $\mathcal{A}$ has only two exterior generators. Both summands are essential to the BF presentation.

To fix the current normalization, define

$$\int^{\text{res}} = \frac{1}{2\pi i} \int$$

on compactly supported complex-valued top forms, both on M and on Σ . The second integral is ordinary oriented integration. In contour notation $\text{Res}_{z=0} z^{-1} dz = 1$. For a compactly supported two-form θ on Σ and a compactly supported function ρ on the half-line, fiber integration has the sign

$$\int_M^{\text{res}} \theta \wedge \rho(t) dt = - \left(\int_0^\infty \rho(t) dt \right) \int_\Sigma^{\text{res}} \theta.$$

Indeed, the orientation of M is the negative of the product orientation on $\Sigma \times \mathbb{R}_{\geq 0}$. Changing the common scalar normalization rescales the action and inverse pairing together.

For homogeneous smooth fields $u = (\alpha, \beta)$ of form degree p and $v = (\alpha', \beta')$, define the local pairing density

$$\omega_{\text{loc}}(u, v) = (-1)^{p+1} (\alpha \wedge \beta' + \beta \wedge \alpha')_{[3]},$$

where $[3]$ takes the top-form component. The orientation identifies this top form with a complex-valued density. Its integral

$$\omega(u, v) = \int_M^{\text{res}} \omega_{\text{loc}}(u, v)$$

is defined whenever at least one argument has compact support. Thus the integrated pairing is defined on $E_{\kappa, c} \otimes E_\kappa$ and on $E_\kappa \otimes E_{\kappa, c}$; no integral is assigned to an arbitrary pair of unrestricted smooth fields. The local pairing has degree -1 , is independent of κ , and is fiberwise nondegenerate: the two complementary summands pair by evaluation and wedge product. Its sign gives graded skew-symmetry. Indeed, a nonzero pairing has $p + q = 3$, so interchanging the forms leaves their wedge sign unchanged and changes $(-1)^{p+1}$ to its negative.

The distinction is already necessary for fields away from the boundary. If χ is a nonzero real smooth function supported in $(1, 2)$, then $u = (\chi(t)dt, 0)$ and $v = (0, \chi(t)dz \wedge d\bar{z})$ have a well-defined local pairing density, but its integral contains the infinite area of $\mathbb{C}$. Their support is not compact in M .

1.2 THE BV IDENTITY AND THE CHIRAL BOUNDARY CONDITION

The square of the differential is zero for every κ :

$$Q_\kappa^2(\alpha, \beta) = (\circ, \kappa(d'\partial + \partial d')\alpha) = \circ.$$

The linear complex is elliptic, including at zero level. For a nonzero real covector ξ , the symbol of d' is wedge multiplication by $\xi^{\circ,1} + \xi_t dt$. This one-form is nonzero: a real covector with zero $(\circ, 1)$ part and zero t part is zero. Contraction with a dual vector contracts this symbol complex. Choose that vector with zero z component. Its contraction anticommutes with the symbol of ∂ , so it also contracts the symbol of Q_κ .

Proposition 1.1 (A nondegenerate BV family). *For every $\kappa \in \mathbb{C}$, the fields, differential and local pairing above define a free BV theory in the interior of M . The boundary condition*

$$L = \Omega^{\bullet}(\Sigma), \quad \iota^* \alpha = \circ, \quad \iota^* \beta \in L, \quad \iota : \Sigma \hookrightarrow M,$$

is a local Lagrangian boundary condition. The restricted differential is compatible with the integrated pairing whenever at least one argument has compact support.

Proof. Ellipticity and nondegeneracy were proved above. For homogeneous u of form degree p , the exterior Leibniz rule gives the local identity

$$\begin{aligned} \omega_{\text{loc}}(Q_\kappa u, v) + (-1)^{p-1} \omega_{\text{loc}}(u, Q_\kappa v) \\ = (-1)^p [d'(\alpha \wedge \beta' + \beta \wedge \alpha') + \kappa \partial(\alpha \wedge \alpha')]_{[3]}. \end{aligned}$$

Assume now that at least one of u, v has compact support. The primitives on the right then have compact support, so their Stokes integrals have no contribution from infinity. The second term integrates to zero: its possible primitive is a two-form generated by $d\bar{z}, dt$, whose boundary pullback vanishes. The first term is an ordinary total derivative, since applying ∂ to its primitive gives holomorphic degree two. Consequently

$$\begin{aligned} \omega(Q_\kappa u, v) + (-1)^{p-1} \omega(u, Q_\kappa v) \\ = (-1)^p \int_{\Sigma}^{\text{res}} \iota^*(\alpha \wedge \beta' + \beta \wedge \alpha'). \end{aligned}$$

Within this support domain the expression vanishes if one argument is supported away from the boundary, or if both satisfy L .

On the boundary, the two Dolbeault summands have equal rank and pair nondegenerately with one another. The second summand is isotropic, so it is Lagrangian. It is preserved by Q_κ because $\partial\beta = \circ$. Its differential is the Dolbeault operator on holomorphic one-forms, an elliptic complex on Σ . The same decomposition tensored with normal forms makes the theory topological in the normal direction. When shifts precede this tensor product, the identification of a normal one-form carries the usual suspension sign; equivalently one may use the exterior-form formula for Q_κ displayed above. Thus the boundary condition is a Lagrangian elliptic subcomplex, and the bulk complex is topological in its normal direction. $\square$

Fix the Hamiltonian convention $\iota_{Q_\kappa} \omega = -\delta S$. The full graded smooth field complex has the local quadratic density

$$S_{\text{loc}}(\Phi) = \frac{1}{2} \omega_{\text{loc}}(\Phi, Q_\kappa \Phi).$$

For $\Phi \in E_{\kappa, c}$ its integral is the well-defined functional $S(\Phi) = \int_M^{\text{res}} S_{\text{loc}}(\Phi) = \frac{1}{2} \omega(\Phi, Q_\kappa \Phi)$. Write $E_{\kappa, L, c} = \{\Phi \in E_{\kappa, c} : \iota^* \alpha = \circ\}$. With the chiral boundary condition imposed, this functional and its

Hamiltonian identity are taken on that space. For unrestricted fields, the action is interpreted as a local density, whose first variation is tested against compactly supported variations.

On ghost-number-zero components the density is

$$\frac{1}{2}(a \wedge d'b + b \wedge d'a + \kappa a \wedge \partial a) = \mathcal{L}_\kappa(a, b) - \frac{1}{2}d'(a \wedge b).$$

For compactly supported fields satisfying $i^*a = 0$, Stokes' theorem kills the total derivative, so its integral is the action stated at the start of the chapter. The local Euler–Lagrange equations, valid on the unrestricted smooth field complex, are

$$d'a = 0, \quad d'b + \kappa \partial a = 0.$$

Their local gauge symmetry, with smooth gauge parameter c vanishing on the boundary, is

$$\delta_c a = d'c, \quad \delta_c b = \kappa \partial c.$$

A compactly supported c also preserves the compactly supported field space. The second equation is invariant because $d'\partial = -\partial d'$.

For any smooth configuration satisfying L and any compactly supported variation $(\delta a, \delta b)$ satisfying the same boundary condition, the first variation of the local action integrates to

$$\begin{aligned} \int_M^{\text{res}} \delta \mathcal{L}_\kappa &= \int_M^{\text{res}} (\delta b \wedge d'a + \delta a \wedge (d'b + \kappa \partial a)) \\ &\quad - \int_\Sigma^{\text{res}} b \wedge \delta a. \end{aligned}$$

All integrands have compact support because each contains a variation or its derivative. Integration by parts is therefore legitimate, and the boundary term vanishes. This proves the variational Hamiltonian identity with its support domain. The configuration $a = t d\bar{z}$, $b = dz$ illustrates the remaining distinction: it satisfies the boundary condition but has a nonintegrable constant action density. Its local variational equations are defined even though it has no integrated action value.

The terms involving ghosts and antifields are fixed by S_{loc} on the unrestricted graded fields. On $E_{\kappa, L, c}$ the integrated functional satisfies the classical master equation: the Hamiltonian vector field of $\{S, S\}$ is $2Q_\kappa^2 = 0$, and this quadratic functional has no constant term. Equivalently, the same statement is an identity of local variational functionals when evaluated against compactly supported variations.

1.3 THE NONZERO-LEVEL CHANGE OF FIELDS AND ITS SPECIAL FIBER

For $\kappa \neq 0$, put

$$F_\kappa(\alpha, \beta) = \alpha + \kappa^{-1}\beta.$$

The target is the full de Rham field complex $\Omega^\bullet(\mathcal{M})[1]$ with differential $d = d' + \partial$ and local pairing density $\omega_{\text{CS}, \kappa, \text{loc}}(u, v) = (-1)^{p+1} \kappa(u \wedge v)_{[3]}$. Its integrated pairing is $\omega_{\text{CS}, \kappa} = \int_M^{\text{res}} \omega_{\text{CS}, \kappa, \text{loc}}$, with at least one compactly supported argument.

Proposition 1.2 (Comparison with abelian Chern–Simons theory). *For every nonzero complex κ , F_κ is an isomorphism of linear BV theories with chiral boundary condition. It identifies the local BV Hamiltonian densities and the integrated BF and level- κ abelian Chern–Simons actions on compactly supported fields satisfying that boundary condition. This field isomorphism has no specialization at $\kappa = 0$.*

Proof. The inverse separates the holomorphic degrees and multiplies the holomorphic-one-form summand by κ . Furthermore,

$$dF_\kappa(\alpha, \beta) = d'\alpha + \partial\alpha + \kappa^{-1}d'\beta = F_\kappa Q_\kappa(\alpha, \beta).$$

The product of two α summands has no three-form component. The product of two β summands has holomorphic degree two and vanishes. The two cross terms consequently give $F_\kappa^* \omega_{\text{CS}, \kappa, \text{loc}} = \omega_{\text{loc}}$ pointwise. Both F_κ and its inverse preserve compact support, so the same identity holds for the integrated pairings on their stated domains. The local Hamiltonian densities, the integrated actions on compactly supported fields, and the boundary condition are preserved. The inverse scalar κ^{-1} in F_κ excludes the zero fiber. $\square$

At zero level the field pairing remains the same nondegenerate cross pairing. In particular, b remains a field, rather than becoming κ times a fixed field. Its gauge transformation vanishes, while the gauge transformation of a remains $d'c$.

This difference can be measured before quantization. On a product of a complex disk D and an interior real interval, the local cohomology of the field complex is computed by

$$O(D) \xrightarrow{\kappa\partial} \Omega_{\text{hol}}^1(D),$$

placed in degrees $-1, 0$. The Dolbeault and real Poincaré homotopies remove the antiholomorphic and real directions. At nonzero κ , holomorphic integration on D makes the arrow surjective and its kernel consists of constants. At zero it is the zero map. Thus

	$H^{-1}(E_\kappa)$	$H^0(E_\kappa)$
$\kappa \neq 0$	$\mathbb{C}$	0
$\kappa = 0$	$O(D)$	$\Omega_{\text{hol}}^1(D)$.

These are cohomology groups of the field complex. The negative-degree summand records gauge data. The zero fiber therefore has more local field cohomology than the nonzero Chern–Simons fibers.

I.4 A GREEN OPERATOR AT ZERO LEVEL

The zero-level kinetic complex has a concrete propagator operator in the interior. Give $\mathbb{C} \times \mathbb{R}$ its flat Euclidean metric and put

$$Q^{\text{GF}} = -4\partial_{\bar{z}}\iota_{d\bar{z}} - \partial_t\iota_{dt}.$$

The contraction symbols satisfy $\iota_{d\bar{z}}(d\bar{z}) = \iota_{dt}(dt) = 1$ and kill the other basis one-forms. Then

$$Q_0 Q^{\text{GF}} + Q^{\text{GF}} Q_0 = -4\partial_z\partial_{\bar{z}} - \partial_t^2 = -\partial_x^2 - \partial_y^2 - \partial_t^2 =: \Delta.$$

The scalar Green kernel is $G(X, Y) = (4\pi|X - Y|)^{-1}$, with $\Delta_X G = \delta_Y$. Consequently $H_0 = Q^{\text{GF}} G$ satisfies

$$Q_0 H_0 + H_0 Q_0 = 1$$

on compactly supported test forms, with output interpreted as smooth forms with the corresponding decay. Its kernel is

$$H_0(X, Y) = \frac{\overline{z - w}}{2\pi|X - Y|^3} \iota_{d\bar{z}} + \frac{t - s}{4\pi|X - Y|^3} \iota_{dt}, \quad X = (z, t), \quad Y = (w, s).$$

The distribution identity follows from the outward flux of $-\nabla(4\pi r)^{-1}$ through a sphere, which is one. Differentiating G gives the two displayed coefficients. No inverse power of κ occurs. Composing with the

inverse cross pairing gives the BV contraction kernel, including the constant $2\pi i$ from the normalization of ω .

This interior operator is not a claim about a boundary Green problem. The boundary comparison below uses compactly supported normal homotopies which preserve the chiral boundary condition. Those homotopies require neither a choice of boundary Green kernel nor a limit of κ^{-1} .

1.5 THE BOUNDARY COMPLEX AND ITS QUANTUM COCYCLE

Write $\mathbf{j}(U) = \Omega_c^{\circ, \bullet}(U)$ for an open $U \subseteq \Sigma$. A generator $u \in \Omega_c^{\circ, q}(U)$ has degree $q - 1$ in the boundary observable complex. The boundary complement is the antiholomorphic summand, while the component of the boundary differential from this complement to L is $\kappa \partial$. Let $\hbar$ be a central polynomial parameter of cohomological degree zero. The boundary observables are

$$\mathcal{B}_\kappa(U) = \left(\text{Sym}(\mathbf{j}(U)[1])[\hbar], \bar{\partial} + \hbar \Delta_{\mu_\kappa} \right), \quad \mu_\kappa(u, v) = -\kappa \int_U^{\text{res}} u \wedge \partial v.$$

The pairing is zero unless the antiholomorphic degrees sum to one. As a contraction of observable generators, it has degree one. The operator Δ_{μ_κ} vanishes on constants and linear terms; on a monomial it sums contractions of unordered pairs, with the Koszul sign for bringing that pair together. This specifies all its values. The differential on a shifted generator is the displayed $\bar{\partial}$, transported without an additional minus sign.

The completion in each symmetric power has a concrete meaning. For the r th tensor power, take compactly supported smooth sections on U^r of the external tensor product of the field bundles; then pass to coinvariants for the graded action of the symmetric group. On every fixed compact support use the seminorms of all derivatives, and take the direct limit over compact supports. This is the smooth kernel, or completed bornological tensor, convention. The symmetric algebra is the direct sum of these powers. Thus polynomial degree is finite; there is no completion in the number of fields.

Lemma 1.3 (Boundary quantum differential). *The square of $\bar{\partial} + \hbar \Delta_{\mu_\kappa}$ vanishes for every κ . At $\kappa = 0$, the quantum boundary complex is exactly the commutative Dolbeault algebra $\text{Sym}(\mathbf{j}(U)[1])[\hbar]$.*

Proof. For u of antiholomorphic degree q , the compatibility equation for one contraction is

$$\mu_\kappa(\bar{\partial} u, v) + (-1)^{q-1} \mu_\kappa(u, \bar{\partial} v) = -\kappa \int_U^{\text{res}} \bar{\partial}(u \wedge \partial v) = 0.$$

Compact support removes the boundary integral. It follows that $\bar{\partial} \Delta_{\mu_\kappa} + \Delta_{\mu_\kappa} \bar{\partial} = 0$ on every symmetric power. In the square of Δ_{μ_κ} , overlapping pairs give zero because a contracted factor has disappeared. Disjoint pairs occur in both orders with opposite signs, since a contraction has odd degree. Thus $\Delta_{\mu_\kappa}^2 = 0$. Finally, $\mu_0 = 0$ by its formula. $\square$

The coefficient κ comes from the differential $\kappa \partial$, not from a degeneration of the field pairing. The next calculation derives this cocycle directly from bulk fields.

1.6 A COMPACT COLLAR MAP FOR QUANTUM OBSERVABLES

Fix a smooth compactly supported function ρ on $\mathbb{R}_{\geq 0}$ with integral one, and put $R(t) = \int_0^t \rho(s) ds$. Thus $R = 1$ for sufficiently large t . Let $E_{\kappa, L, c}(U \times \mathbb{R}_{\geq 0})$ denote compactly supported fields satisfying $\iota^* \alpha = 0$. For $u \in \Omega_c^{\circ, q}(U)$ define

$$I_\kappa(u) = u \wedge \rho(t) dt + (-1)^q (R(t) - 1) \kappa \partial u.$$

This is a degree-zero map from the unshifted complex $(\mathfrak{j}(U), \bar{\partial})$ to $(E_{\kappa, L, c}(U \times \mathbb{R}_{\geq 0}), Q_{\kappa})$. The first term is in the α summand and the second is in the β summand. Both have compact support. The first has zero boundary pullback, and the second lies in L .

The bulk observable generators are sx , for $x \in E_{\kappa, L, c}$, with $|sx| = |x| - 1$. Define their contraction by

$$B(sx, sy) = (-1)^{|x|} \omega(x, y).$$

For complementary form degrees this is the normalized integral of the two cross terms. Let Δ_B sum pair contractions as above. The bulk–boundary quantum complex is

$$\mathcal{O}_{\kappa}(U \times \mathbb{R}_{\geq 0}) = (\text{Sym}(E_{\kappa, L, c}[1])[\hbar], Q_{\kappa} + \hbar \Delta_B).$$

The shifted unary operator is $Q_{\kappa}(sx) = sQ_{\kappa}x$. The pairing identity in Proposition 1.1 and the same disjoint-pair cancellation show that this differential squares to zero.

Theorem 1.4 (The quantum boundary map at every level). *The maps I_{κ} induce quasi-isomorphisms*

$$\text{Sym}(I_{\kappa}) : \mathcal{B}_{\kappa}(U) \longrightarrow \mathcal{O}_{\kappa}(U \times \mathbb{R}_{\geq 0})$$

for every open $U \subseteq \Sigma$ and every complex κ , including zero. They commute with extension by zero and with factorization products on disjoint opens.

Proof. First check the unary differential. Differentiating the first term of $I_{\kappa}(u)$ gives $\bar{\partial}u \wedge \rho dt + \kappa \partial u \wedge \rho dt$. Differentiating the second gives

$$-(-1)^q (R - 1) \kappa \partial \bar{\partial} u - \kappa \partial u \wedge \rho dt.$$

The last terms cancel, leaving $I_{\kappa}(\bar{\partial}u)$.

Projection to the α summand followed by integration along the normal one-form gives a cochain map P , with $PI_{\kappa} = 1$. To prove that P is a quasi-isomorphism, consider the exact sequence whose subcomplex is the β summand and whose quotient is the α summand. The former has absolute normal forms; the latter has relative normal forms, whose functions vanish at $t = 0$. On the normal complexes the homotopies are

$$\begin{aligned} h_{\text{abs}}(g dt) &= - \int_t^{\infty} g(s) ds, \\ h_{\text{rel}}(g dt) &= \int_0^t g(s) ds - R(t) \int_0^{\infty} g(s) ds. \end{aligned}$$

Both vanish on functions. The absolute homotopy satisfies $dh + hd = 1$. The relative one satisfies $dh + hd = 1 - ip$, where $p(g dt) = \int g$ and $i(1) = \rho dt$ in degree one. Their values are compactly supported, and the relative value vanishes at $t = 0$. Direct differentiation proves all these identities. For a coefficient of tangential form degree r , multiply the normal homotopy by $(-1)^r$; this cancels the tangential cross terms. These formulas also apply to smooth sections depending on z . The β subcomplex is therefore acyclic, and the relative α quotient retracts onto $\mathfrak{j}(U)$ after its displayed shift. The long exact sequence proves that P , hence I_{κ} , is a quasi-isomorphism.

There is also a continuous contraction for the full split complex. Let H be the two normal homotopies just displayed, and put $\partial_{\kappa} = Q_{\kappa} - Q_0$. The correction goes from the α summand to the β summand, so $\partial_{\kappa} H \partial_{\kappa} = 0$ and $P \partial_{\kappa} = 0$. Consequently

$$H_{\kappa} = H - H \partial_{\kappa} H, \quad I_{\kappa} = I_0 - H \partial_{\kappa} I_0, \quad Q_{\kappa} H_{\kappa} + H_{\kappa} Q_{\kappa} = 1 - I_{\kappa} P.$$

To check the last identity, substitute $Q_0 H + H Q_0 = 1 - I_0 P$ and $Q_0 \partial_{\kappa} = -\partial_{\kappa} Q_0$. The linear correction terms cancel except for $H \partial_{\kappa} I_0 P$; the quadratic correction terms vanish. The formula for I_{κ} is the displayed collar map, since the absolute homotopy of ρdt is $R - 1$. This construction uses no division by κ .

It remains to check the quantum contraction, rather than only the unary differential. For antiholomorphic degrees $q + r = 1$, expand the two cross terms in $I_\kappa(u) \wedge I_\kappa(v)$. Moving dt to the right, and integrating by parts on U , gives

$$B(sI_\kappa(u), sI_\kappa(v)) = -2\kappa \left(\int_0^\infty \rho(t)(1 - R(t))dt \right) \int_U^{\text{res}} u \wedge \partial v.$$

The minus sign is the fiber-integration sign fixed above; B itself integrates the two cross terms with positive sign. For the second term use $\int_U \partial u \wedge v = (-1)^{q+1} \int_U u \wedge \partial v$. The normal integral is

$$\int_0^\infty \rho(1 - R)dt = \left[R - \frac{R^2}{2} \right]_0^\infty = \frac{1}{2}.$$

Consequently $B(I_\kappa(u), I_\kappa(v)) = \mu_\kappa(u, v)$, with suspensions understood. This proves compatibility on quadratic monomials. The pair-contraction definitions prove it on every symmetric power.

At the classical level, the preceding continuous linear homotopies extend to r tensor factors by the explicit map

$$\mathcal{H}_r = \sum_{j=1}^r (I_\kappa P)^{\otimes(j-1)} \otimes H_\kappa \otimes 1^{\otimes(r-j)}.$$

Use the Koszul sign when the degree-minus-one map H_κ passes the preceding factors. In $d\mathcal{H}_r + \mathcal{H}_r d$, the terms $1 - I_\kappa P$ telescope, giving $1 - (I_\kappa P)^{\otimes r}$. Averaging over the finite symmetric group gives the result on symmetric powers; characteristic zero permits this averaging. The integrations preserve compact supports in the product variables and extend to the completed bornological tensors. Thus $\text{Sym}(I_\kappa)$ is a classical quasi-isomorphism.

Finally filter both polynomial quantum complexes by the sum of the power of $\hbar$ and the number of fields. The unary differential preserves that filtration degree. A quantum contraction reduces it by one. The associated graded map is the classical quasi-isomorphism tensored with $\mathbb{C}[\hbar]$. The filtration starts in degree zero, and every element has finite filtration degree. Induction on the finite filtration, followed by the exact direct limit, proves the quantum quasi-isomorphism. All formulas use the same ρ and integrate only the normal variable. They therefore commute with extension by zero and with products on disjoint boundary opens. $\square$

The observables also satisfy the gluing property needed for a factorization algebra. The relevant covers $\{U_i\}$ of U are those for which every finite subset of U lies in some U_i . For every r , the sets U_i^r then cover U^r . At a fixed field degree the augmented Čech complex of compactly supported smooth kernels for this cover is exact: a locally finite partition of unity gives its contraction, and only finitely many terms meet a given compact support. Multiplication by the partition functions preserves the linear boundary condition for bulk fields. For fixed r the form degrees are bounded, so the finite filtration by form degree upgrades this Čech exactness to descent for the unary differential. Passing to symmetric coinvariants preserves exactness because averaging over the finite symmetric group is available. The polynomial filtration used above then proves quantum descent. Here factorization algebra means precisely these gluing complexes together with the compatible products for disjoint opens. Thus the theorem is an equivalence of quantum factorization algebras on the boundary. At $\kappa = 0$ its image contains only α fields, so its pair contraction vanishes directly.

1.7 THE POLYNOMIAL BOUNDARY CURRENT AT ZERO LEVEL

At $\kappa = 0$, the boundary differential is $\bar{\partial}$ and there are no quantum pair contractions. The current states can be read from compactly supported Dolbeault forms on a disk $D = \{|z| < R\}$ about zero. Choose a smooth radial cutoff χ , equal to one near zero and zero near the boundary of D . For $n \geq 0$ set

$$u_n = -\frac{\bar{\partial}\chi}{z^{n+1}} \in \Omega_c^{\alpha,1}(D).$$

Its support avoids zero, so the form is smooth. Stokes' theorem on an annulus gives, for every holomorphic f on D ,

$$\int_D^{\text{res}} u_n \wedge f(z) dz = \text{Res}_{z=0} \frac{f(z) dz}{z^{n+1}} = \frac{f^{(n)}(0)}{n!}.$$

Thus u_n is the divided-jet evaluation functional.

For completeness, the one-variable compact-support calculation has no hidden odd class. A compactly supported holomorphic function is zero. A compactly supported $(0, 1)$ -form u is $\bar{\partial}$ -exact with compact support if all its holomorphic moments vanish. Indeed, its Cauchy transform

$$(Tu)(z) = \frac{1}{2\pi i} \int_D \frac{u(w) \wedge dw}{z - w}$$

satisfies $\bar{\partial}Tu = u$: Stokes' theorem on a small circle gives the distribution identity $\partial_{\bar{z}}(1/(\pi z)) = \delta_0$, and $d\bar{w} \wedge dw = 2i dx \wedge dy$ fixes the factor in T . Outside the support disk, expansion in w/z expresses Tu through those moments. If they all vanish, Tu is zero there and hence has compact support in D . Conversely, integration by parts kills the moments of $\bar{\partial}f$ for compactly supported f . For a fixed rotation weight, there is at most one nonzero holomorphic moment. The classes $[u_n]$ are consequently a basis of the rotation-finite part of $H_c^{0,1}(D)$.

The same calculation controls the needed tensor completion. If u is supported in $|z| \leq r < R$, choose the cutoff annulus at radii larger than r . Its moments satisfy $|M_n(u)| \leq Cr^n$. On an annulus with inner radius $r' > r$, every fixed derivative of u_n is bounded by $C_d(n+1)^d (r')^{-n-1}$. Thus $\sum_n M_n(u)u_n$ converges with every derivative and has compact support. Subtracting it removes all moments, and the Cauchy transform gives a compact primitive. These maps are continuous on each fixed compact-support space. A finite set of inputs has a common such space and cutoff.

On a product of r disks, apply this compact-support Dolbeault calculation in each variable. The completed tensor powers are compactly supported forms on that product. Iterated Cauchy integration and the tensor homotopy compute their finite joint-rotation-weight components. The radial cutoffs and Cauchy homotopies commute with rotations, so the same contraction restricts to a fixed diagonal rotation weight; it does not require a finite Fourier expansion in each variable. Their cohomology is in top antiholomorphic degree r , with surviving tensors $[u_{n_1}] \otimes \cdots \otimes [u_{n_r}]$. The conformal weight is minus the pullback rotation weight on test forms. For total conformal weight N , only $\sum_i (n_i + 1) = N$ survives, so there are finitely many such tensors. Passing to symmetric coinvariants is exact over $\mathbb{C}$. It follows that the polynomial, finite-weight vacuum states of the boundary observable algebra are

$$V_o = \mathbb{C}[q_1, q_2, \dots][\hbar], \quad q_{n+1} = [u_n], \quad |q_m| = 0, \quad \text{wt}(q_m) = m.$$

The ghost-degree-minus-one generators in the Dolbeault model have not been discarded: their differential is part of the calculation which leaves these degree-zero classes.

Moving the evaluation point from zero to z gives the translation operator

$$Tq_m = mq_{m+1}, \quad Y(a, z)b = (e^{zT}a)b \quad (a, b \in V_o).$$

Here T extends as a derivation. For a field $Y(a, z)$, its mode $a_{(n)}$ is the coefficient of z^{-n-1} . The exponential is a formal series in z , and each coefficient is a well-defined polynomial. The field of q_1 is therefore

$$J(z) = Y(q_1, z) = \sum_{m \geq 1} q_m z^{m-1}.$$

All its nonnegative modes vanish, and J_{-m} acts by multiplication by q_m . Multiplication of compactly supported boundary observables on disjoint disks gives the product of the evaluation functionals. The moving-point formula therefore gives the displayed vertex product, including its regular terms, rather than only its singular part.

Theorem 1.5 (The zero-level current comparison). *The polynomial chiral vacuum extracted from the boundary observables of zero-level holomorphic BF theory is the commutative differential vertex algebra $\mathbb{C}[q_1, q_2, \dots][\hbar]$ with $Tq_m = mq_{m+1}$ and $Y(a, z)b = (e^{zT}a)b$. After coefficientwise $\hbar$ completion, it is isomorphic to the algebraic Heisenberg vacuum specialized at $\kappa = 0$. The isomorphism preserves every mode, holomorphic jet, regular product and separated field product.*

Proof. The compact-support calculation gives the state space and its degree. Moving the jet functional gives T , and factorization gives the product just calculated. These formulas satisfy the vacuum, translation and associativity identities because multiplication is commutative and associative and T is a derivation.

For the algebraic comparison, the Heisenberg vacuum over $\mathbb{C}[\kappa][\hbar]$ has mode action

$$J_{-m} = q_m, \quad J_m = \hbar \kappa m \partial_{q_m} \quad (m > 0), \quad J_0 = 0.$$

At zero level this is precisely the action above, including all positive modes and the vacuum. The map $q_{n+1} \mapsto [u_n]$ is bijective on the monomial basis. It intertwines T and every field, so it preserves regular and separated products as stated. All formulas are polynomial in the generators and in $\hbar$. Completing their coefficients therefore gives the same comparison without completing the number of fields in an individual coefficient. $\square$

The current-mode normalization can also be computed on an annulus. Let f, g be holomorphic Laurent polynomials there, and choose χ equal to one near its inner boundary and zero near its outer boundary. The annular transgression evaluates the local cocycle on the holomorphic lift f and the compactly supported form $\delta\chi g$. Integration by parts gives

$$\begin{aligned} \mu_\kappa(f, \delta\chi g) &= -\kappa \int_{\Sigma}^{\text{res}} \delta\chi \wedge g df \\ &= \kappa \text{Res } g df = -\kappa \text{Res } f dg. \end{aligned}$$

Only one factor needs compact support for this integral. The inward boundary of the annulus supplies the minus sign in $\int_{\Sigma}^{\text{res}} \delta\chi \wedge g df = -\text{Res } g df$. This is the contour cocycle for current modes. On Laurent monomials,

$$\kappa \text{Res } g df = \kappa m \delta_{m+n,0} \quad (f = z^m, g = z^n).$$

Thus the boundary quantum differential has the contour current cocycle $\kappa \text{Res } g df$. More explicitly, a central extension of Laurent polynomials is obtained by adjoining a central symbol and using this bilinear expression as the bracket. It is antisymmetric because $\text{Res } d(fg) = 0$; its Jacobi identity holds because every bracket is central. The mode representation above realizes its quantum normalization directly:

$$[q_m, \hbar \kappa n \partial_{q_n}] = -\hbar \kappa n \delta_{m,n}, \quad [J_m, J_n] = \hbar \kappa m \delta_{m+n,0}.$$

Write $J_+(z) = \sum_{m>0} J_m z^{-m-1}$ and $J_-(w) = \sum_{m>0} J_{-m} w^{m-1}$. For $|z| > |w|$, the positive-negative contraction is

$$[J_+(z), J_-(w)] = \hbar \kappa \sum_{m \geq 1} m z^{-m-1} w^{m-1} = \frac{\hbar \kappa}{(z-w)^2}.$$

This proves agreement of the actual local quantum cocycle with the scalar and mode normalization. The full extraction of boundary fields proved here is the zero-level extraction in Theorem 1.5. The

normalization is thus continuous in the split family, although the nonzero-level change of fields is singular at zero. At zero level the mode cocycle vanishes directly, as does Δ_{μ_0} ; no limiting inverse pairing was used.

Corollary 1.6 (Augmentation of the extracted current). *The zero-level polynomial vertex algebra has the augmentation $q_m \mapsto 0$ for all $m \geq 1$. The corresponding nonzero-level Heisenberg vertex algebra has no augmentation to $\mathbb{C}[\hbar]$ as a vertex algebra over $\mathbb{C}[\hbar]$.*

Proof. At zero level, evaluation at all $q_m = 0$ preserves multiplication, translation and the vacuum. At nonzero level, $J_{(1)}J = \hbar\kappa 1$. Any vertex map to the scalar vertex algebra would send the left side to zero and the right side to the nonzero scalar $\hbar\kappa$. This is impossible. $\square$

1.8 SEPARATED PRODUCTS AND TOPOLOGICAL HIGHER OPERATIONS

For distinct holomorphic insertion points, the zero-level Wick product has no contraction terms. Its normal-ordering map sends

$$x_{i,r} \mapsto \partial_{z_i}^r J(z_i), \quad x_{1,0} \star x_{2,0} \star x_{3,0} = x_{1,0}x_{2,0}x_{3,0}.$$

These are equalities on the actual boundary field representation from Theorem 1.5.

An auxiliary topological interval gives the same higher operations on these fields as on their algebraic presentation. To specify the statement, let e_0, e_1 be endpoint cochains and η the oriented edge, with $de_0 = -\eta$, $de_1 = \eta$. Their polynomial-form inclusion and homotopy are

$$i(e_0) = 1 - t, \quad i(e_1) = t, \quad i(\eta) = dt, \quad h(gdt) = \int_0^t g - t \int_0^1 g.$$

Projection takes endpoint values and the integral of a one-form. Direct differentiation gives $dh + hd = 1 - ip$ and $hi = ph = h^2 = 0$. Extend these maps by the identity on boundary field coefficients.

To check all higher identities, let W be the interval cochains with these field coefficients and let A be their polynomial-form resolution. The coefficient algebra is the commutative algebra of represented zero-level current fields at the chosen separated points, interpreted coefficientwise in the insertion variables. Write sa for a suspension, with $|sa| = |a| - 1$. Put

$$D_1(sa) = s(da), \quad D_2(sa, sb) = (-1)^{|a|} s(ab).$$

Extend these maps to the direct sum of finite tensor words by inserting them into each consecutive block, with the Koszul sign from the preceding suspended inputs. This is a coderivation for the coproduct which splits words into initial and final segments. Associativity and the Leibniz rule say $(D_1 + D_2)^2 = 0$. Use the same letters i, p, h for the suspended linear maps and define

$$F_1 = i, \quad R_n = \sum_{a+b=n} D_2(F_a \otimes F_b), \quad F_n = -hR_n, \quad b_n = pR_n \quad (n \geq 2),$$

with $b_1 = sds^{-1}$. An A_∞ structure here means that the resulting degree-one coderivation b on the tensor words in sW satisfies $b^2 = 0$. The ordinary operations are fixed by

$$b_n(sa_1, \dots, sa_n) = (-1)^{\sum_j (n-j)|a_j|} sm_n(a_1, \dots, a_n).$$

Here is a direct verification of the recursion. The maps F_n extend uniquely to a coalgebra map F . Induct on word length, assuming $DF = Fb$ and $b^2 = 0$ in smaller lengths, where $D = D_1 + D_2$. With the length- n terms temporarily zero, the length- n morphism defect is

$$Q_n = R_n - L_n, \quad L_n = \sum_{\substack{r+q+t=n \\ 2 \leq q \leq n-1}} F_{r+1+t}(1^{\otimes r} \otimes b_q \otimes 1^{\otimes t}).$$

The tensor substitutions carry the insertion signs just specified. Let O_n be the length- n term of b^2 with $b_n = 0$. The identity

$$D(DF - Fb) + (DF - Fb)b = -Fb^2$$

gives $\delta Q_n = -iO_n$, where $\delta Q_n = D_1 Q_n + Q_n b_1$ and b_1 also denotes its extension to words. For $k \geq 2$, $pF_k = hF_k = 0$, by $ph = h^2 = 0$. It follows that $pL_n = hL_n = 0$. Setting $b_n = pQ_n = pR_n$ gives $\delta b_n = -O_n$, which is the missing length- n identity for b^2 . Setting $F_n = -hQ_n = -hR_n$ and using $D_1 h + hD_1 = 1 - ip$ gives

$$\delta F_n = -Q_n + ipQ_n + h\delta Q_n = ib_n - Q_n,$$

because $hi = 0$. This is the missing morphism identity. The induction starts with the cochain maps i, p , so it proves every arity without an infinite sum on any word. The linear map i is a quasi-isomorphism by the explicit contraction.

Every field comparison intertwines each multiplication vertex and each interval homotopy separately. It therefore intertwines these operations in every arity, before summing their binary trees. For example,

$$h(tdt) = \frac{t^2 - t}{2}, \quad \int_0^1 \frac{t - t^2}{2} dt = \frac{1}{12}$$

gives, in ordinary cohomological degrees $|m_n| = 2 - n$,

$$m_3(e_1 J(z_1), \eta J(z_2), \eta J(z_3)) = -\frac{\eta}{12} J(z_1) J(z_2) J(z_3).$$

The degree-one output reflects the interval cochains. It is compatible with the strict degree-zero vertex algebra because this interval resolution has nonzero differential. It is not a cubic interaction in the BF action.

The quantum boundary map above identifies actual supported observables on the boundary with their bulk–boundary pushforward. The current comparison identifies its polynomial chiral vacuum, and the interval argument identifies the transferred separated products. These three maps have different sources. In particular, the last map alone does not describe a bulk action, a collision-chain compactification, or higher operations of an interacting theory. The zero-level BF construction supplies a physical realization of the zero-level current while retaining the additional bulk gauge and field cohomology computed above.